

Paradigm

- Inspiration for a conceptual model
- General approach adopted by a community for carrying out research
 - shared assumptions, concepts, values, and practices
 - e.g. desktop, ubiquitous computing, in the wild

Examples of new paradigms

- Ubiquitous computing (mother of them all)
- Pervasive computing
- Wearable computing
- Tangible bits, augmented reality
- Attentive environments
- Transparent computing
 - and many more....

Visions

- A driving force that frames research and development
- Invites people to imagine what life will be like in 10, 15 or 20 years time
 - e.g. Apple's 1987 Knowledge Navigator
 - Smart Cities, Smart Health
- Provide concrete scenarios of how society can use the next generation of imagined technologies
- Also raise many questions concerning privacy and trust

Theory

- Explanation of a phenomenon
 - e.g. information processing that explains how the mind, or some aspect of it, is assumed to work
- Can help identify factors
 - e.g. cognitive, social, and affective, relevant to the design and evaluation of interactive products

Models

- A simplification of an HCI phenomenon
 - intended to make it easier for designers to predict and evaluate alternative designs
 - abstracted from a theory coming from a contributing discipline, e.g. psychology, e.g. keystroke model

Framework

- Set of interrelated concepts and/or specific questions for 'what to look for'
- Many in interaction design
 - e.g. Norman's conceptual models, Benford's trajectories
- Provide advice on how to design
 - e.g. steps, questions, concepts, challenges, principles, tactics and dimensions

Concern	Past	Future
Frame of reference	• users	• context
Method, theory, and perspective	• scientific approach • interaction design	• pluralistic • mixing
Outputs	• ethnographies • models and tools for analysis • design guidance	• insights • creating new ways of experiencing • value-based analyses

Table 2.1 A new framework for human–computer interaction (Rogers, 2009)